

CHAPTER IX

Outside Drivers

"The only discipline, important for itself, is self-discipline, and this can only be acquired by a wide use of freedom."

Alfred North Whitehead, "The Rhythmic Claims of Freedom and Discipline"

The Reading Chart

In June, the library posts a summer reading chart. Every book read earns a sticker, and every ten stickers earn a coupon for a free pizza.

A ten-year-old girl takes the chart seriously. By the end of July she has forty stickers and four pizzas. She chose the thinnest books on the shelf, and she can tell you very little about any of them. On the first of September, when the chart comes down, she stops reading. By October she hasn't opened a book that wasn't assigned.

Her younger brother ignored the chart. In August he found a long fantasy novel in a box in the garage, and he read it under the covers with a flashlight, then read the sequel, then went looking for the third. Nobody gave him a sticker. He's still reading.

The chart worked, in a sense. It got the girl reading forty books, which is a lot of books. And the moment it came down, there was nothing left.

This chapter is about drivers like the chart: the engagement drivers that come from outside the learning object. They come in three families: **social** (competition, cooperation, an audience, models, belonging, praise), **story** (narrative, curiosity, play, entertainment), and **structural** (rewards, grades, money, progress, streaks, leaderboards, deadlines, games).[1] People are social, imitative, and playful, and a model of learning that relies on solitary curiosity alone leaves its strongest drivers unused.

Outside drivers capture attention. Engagement is something narrower: sustained mental investment in the thing being learned.[2] Capture can turn into engagement, and much good learning starts that way. An outside driver is judged by where it leads: toward the learner owning the work, or away from the object and onto the driver itself.[3] No outside driver is corrupt in itself. Each becomes corrupting when it pulls attention off the object and holds it there.

Chapter 8 covered ownership at every scale, including seeding it, identity labels, partnership, and the stories learners tell. This chapter points back to that material where it can, and spends its space on the drivers that begin outside ownership.[4]

The Same Driver, Two Ways

As in the last two chapters, the clearest way to see the difference is to hold one thing still. Here it's the driver. In each pair, the driver is the same. Only where it points changes.

Competition: The Spelling Bee

	CAPTURE	ENGAGEMENT
What the learner watches	The classmate she wants to beat	The words: the harder, the better

	CAPTURE	ENGAGEMENT
What a win means	She came out on top	She can spell words she couldn't spell last month
What a loss does	Ends her interest in spelling	Sends her to the dictionary, to the word she missed
After the contest	Spelling stops until next year's bee	She keeps a list of words she wants to master
What lasts	A ranking	A vocabulary, and a way to build it

Story: The Dinosaur

	CAPTURE	ENGAGEMENT
Where the story points	A cartoon dinosaur dances between math drills as a reward	The class is a dig team, and measurement tells them how big the dinosaur was
What the math is for	Getting through to the next cartoon	Answering a question the story raised
What the learner remembers	The cartoon	How long a femur has to be to hold up twelve tons
When the story ends	The math ends with it	The math keeps its meaning

Structure: The Reading Streak

	CAPTURE	ENGAGEMENT
What the streak counts	One page read before midnight to keep the count alive	Nights spent in a book she can't put down
What the learner watches	The number	The story
When the streak breaks	Reading stops; the count was the point	She picks the book back up the next night
What lasts	A record	A reader

In every pair, the driver does its job: it gets the learner started. The difference is whether it points at the object.

Think of a turn signal and a steering wheel. A driver that points at the object works like a turn signal: it gets your attention, and then the wheel takes over and the car actually turns. A driver that points at itself is a turn signal that never leads to a turn. It has to keep blinking, brighter and faster, to hold you. Today the reward is a dancing dinosaur; next month it has to be a dragon; next year, a whole video game.

A driver that points at the object can hand its work over to the object. A driver that points at itself has to keep working forever.

❧ Capture and Engagement

Two words, then. **Capture** is attention held by a driver: the streak count, the rank, the reward, the cartoon. **Engagement** is mental investment in the learning object itself. A learner can be fully captured and barely engaged; the girl with forty stickers was. The central question of this chapter is when capture turns into engagement.

The Continuum

Richard Ryan and Edward Deci (2000) describe motivation along a continuum of **internalization**: a road that runs from outside the learner to inside. Take someone who goes to the gym, and walk her down it.

- **External regulation.** Acting to get a reward or avoid a punishment. She goes because her insurer gives her a discount for three visits a week.
- **Introjected regulation.** Acting to avoid guilt or to win approval, including her own. She goes because she'd feel terrible if she didn't, or because she wants to be seen as disciplined.
- **Identified regulation.** Acting because she values what the activity leads to. She doesn't love the treadmill, but she values the stamina it gives her to keep up with her kids.
- **Integrated regulation.** Acting because the activity has become part of who she is. She's a person who trains.

- **Intrinsic motivation.** Acting because the activity itself is interesting or satisfying. She loves the feeling of lifting.

From outside, all five look the same: a woman lifting a weight. Motivation that starts outside the learner can move along this road, and most outside drivers begin at the external end.

What Each Point on the Road Produces

Does the starting point matter if the weight gets lifted either way? The evidence says it matters a great deal.

Joshua Howard, Julien Bureau, Frédéric Guay, Jane Chong, and Richard Ryan (2021) pooled 344 samples and more than 223,000 students. Intrinsic motivation was related to school success and to well-being. Identified regulation, valuing what the activity leads to, was especially strongly related to persistence. Introjected motivation, acting from guilt or for approval, predicted persistence too, but it also predicted signs of ill-being: people kept showing up and felt worse for it. External regulation, acting for rewards or to avoid punishment, wasn't associated with performance or persistence at all, and it was associated with lower well-being.

Quality and Quantity

Christopher Cerasoli, Jessica Nicklin, and Michael Ford (2014) looked across forty years of research in school, work, and sport: 183 samples, more than 212,000 people. Intrinsic motivation predicted performance moderately to strongly whether or not incentives were present. And the two did different jobs. Intrinsic motivation better predicted the **quality** of performance. Incentives better predicted its **quantity**.

Outside drivers can buy more reps. They don't buy better ones. Forty thin books is exactly what a sticker chart should be expected to produce.

The Standard

An outside driver is judged by whether it moves the learner along the continuum toward owning the work, or holds them at the external end.

Capture that stays capture is what the evidence warns against. Capture that hands the learner over to the object is how much good learning starts.

At the far end of the road, the learner does the work because it has become theirs: they care about the questions, and they hold the craft to their own standard. That's engagement that no longer needs an outside driver. Chapter 4 argued that sustained engagement becomes care and craft. The continuum is the path it takes.

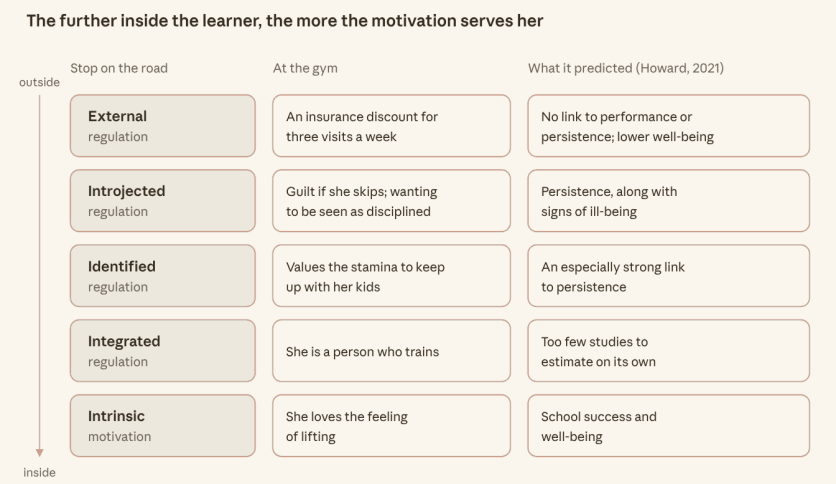


Figure 9.1 · the internalization continuum (schematic)

From outside, all five stops look like a woman lifting a weight. Further along the road, the motivation predicts more of what lasts.

> ***Box: Mimetic Desire***
>
> *Why are we so drawn to outside drivers in the first place? René Girard (1923–2015), a literary critic and anthropologist, offers an answer. He argued that human desire is* ***mimetic****. People rarely want things on their own; they want what their models want. Desire runs in a triangle, from the learner through a model to the object. Don Quixote sets out to be a knight because the hero of his books, Amadis of Gaul,*

was one (Girard, Deceit, Desire, and the Novel, 1961). We like to think our desires are our own. Girard thought most of them are borrowed.

Two kinds of model. *Girard distinguished* ***external mediation****, where the model stands outside the learner's world (a hero, a historical figure, an adult far ahead), from* ***internal mediation****, where the model stands inside it: a peer who wants the same thing. An external model can be imitated without rivalry; you can want to do physics the way Einstein did without ever competing with him for a grant. An internal model becomes a rival, and in rivalry the object tends to disappear. The rivals come to want each other's standing, and the thing itself stops mattering. Girard called desire for another's prestige or status* ***metaphysical desire****.*

The evidence. *Girard built his theory from novels, myth, and history, and parts of it, especially his account of scapegoating and religion, remain contested. The core claim about imitative desire has experimental support. People rated objects as more desirable after seeing them as the goal of another person's action, and the effect ran through the brain systems for observing action and assigning value (Maël Lebreton and colleagues, 2012). The research on prestige-biased learning later in this chapter points the same way.*

A filter for the outside drivers. *Mimetic theory sorts the drivers by where they place the model.*

- Admired adults, heroes, and older learners are external models. Their desire for the object can pass to the learner without rivalry.
- Task-oriented competition aims at a standard. Other-referenced competition and leaderboards turn peers into internal models, and invite the object to disappear.
- Grades, ranks, and badges are metaphysical objects: they stand for standing among peers.
- Streaks, likes, and public counts make other people's attention the object.

- A teacher who visibly loves the work passes on desire for the work. A teacher who mainly cares about scores passes on desire for scores.

The design rule. *People will imitate; the question is whom. Choose models whose desire points at the object, and keep them at a distance that allows admiration without rivalry.*

A limit. *Pushed too far, Girard's theory makes every desire look borrowed and every influence corrupting. It works better as a lens than as a law. A young person who becomes a nurse because she admired the nurse who cared for her grandmother is imitating, and may be doing exactly the right thing.*

The school through Girard's lens. *The classroom form is close to an ideal setting for internal mediation.*[5]

- **Every model is a peer.** Children spend most of the school day with others of the same age, at the same level, doing the same tasks, measured by the same standard. The one adult in the room is outnumbered thirty to one. Older students are in other rooms; working adults and experts are outside the building. Nearly every model a child can see is too close to admire from a distance, and close enough to become a rival.
- **Every peer wants the same things.** The grade, the rank, the teacher's approval, the place in the next school. These are metaphysical objects: they stand for standing. When learning becomes the way to get them, the subject becomes a means, and the object disappears between the rivals.
- **The arrangement is recent.** Age-graded classrooms became common only in the nineteenth century. For most of human history, children learned among people of many ages, in families, workshops, and communities, where many of the models were far enough ahead to admire.

- **The evidence in this chapter fits.** Children learn most readily from admired models. Cooperation outperforms individual competition, and teams competing against other teams outperform individuals competing against each other: a shared rival outside the group turns internal models back into partners. Task-oriented competitors are well liked; other-referenced competitors aren't. Each finding is what the theory predicts. Rivalry among near equals costs, and admiration across distance pays.
- **What the lens recommends.** Supply external models: older learners, adults who love the work, masters and apprentices, tutors, heroes from the past.[6] Aim competition at standards and at other teams, and away from the classmate at the next desk. Replace ranks with comments. The classroom form can soften internal mediation. It can't escape it: same-age peers ranked against each other is what the form is.

❧ Social Drivers

Each driver in the next three sections gets the same treatment: what the research shows, when the driver points at the object and when at itself, how it changes with age, and how to design with it.

Competition

If you're a competitive person, you may be bracing. Sometimes the only way you get through a dull report is by racing a colleague to finish it first.

The broad evidence favors cooperation. Across 148 studies of more than 17,000 early adolescents in 11 countries, cooperative goal structures produced higher achievement than competitive ones (an effect size of 0.46) and individualistic ones (0.55), along with better peer relationships (Cary Roseth, David Johnson, and Roger Johnson, 2008). An earlier meta-analysis found that teams competing against other teams outperformed individuals

competing against each other (Johnson and colleagues, 1981). In a classic experiment, adults who solved puzzles while trying to beat another person showed less intrinsic motivation afterward than adults simply trying to finish quickly (Edward Deci and colleagues, 1981). And the outcome matters: winning tends to raise later interest in a task, and losing tends to lower it (Johnmarshall Reeve, Bradley Olson, and Steven Cole, 1985).

But "competition" covers two different things. **Task-oriented competition**, competing to improve your own performance, goes with being well liked and seen as prosocial. **Other-referenced competition**, competing mainly to beat an opponent, goes with being liked less and seen as aggressive (Fabrizia Tassi and Barry Schneider, 1997). Racing a friend so you both run a faster mile is the first kind. Not caring how slow you run as long as he runs slower is the second. The spelling bee at the start of the chapter is this distinction.

When it points at the object. Racing a standard, a personal record, or another team keeps the task in view. Ranking individuals against each other puts the rival in view, and the task recedes.

With age. Differences in willingness to compete appear early. In an experiment with 1,035 children aged three to eighteen, boys chose to enter competitions far more often than girls, from age three on (Matthias Sutter and Daniela Glätzle-Rützler, 2015). In a math game for 274 first and second graders, competitive and collaborative play both boosted perseverance and attitudes compared with playing alone, but only competition produced gender differences in attention to the math (Margaret Fish and colleagues, 2023). A purely competitive design can push half a room's attention off the work.

Design moves. Compete against a standard or a past self. Compete as teams. Make losing cheap and frequent, so a loss is information. Offer competition as a choice.

Cooperation

Cooperation's advantage depends on structure, as anyone who has suffered a group project knows: one person does the work, two coast, one brings snacks. Robert Slavin's reviews (1995) find that cooperative learning raises achievement when a group shares a goal **and** each member is individually accountable, so the team succeeds only if every member learns. Methods built on team-building alone did no better than traditional instruction, and the original jigsaw method hasn't generally raised achievement. Telling students to work together falls short. They need a reason to take one another's learning seriously.

The feeling of togetherness alone helps. In two studies with 70 preschoolers, merely signaling that a child was working on a puzzle together with someone else, with no actual help given, substantially increased persistence on the puzzle and liking for it (Lauren Butler and Gregory Walton, 2013).

When it points at the object. The shared goal is the work itself. Cooperation points away from the object when the group becomes the goal: when fitting in or avoiding friction matters more than the task, or when one member quietly does the work for all.

With age. The drive to share goals appears in toddlerhood. Structured roles matter more as tasks grow complex. Partnership and group ownership are developed in Chapter 8.[7]

Design moves. Give the group one real goal. Make each member's contribution visible. Rotate roles. Keep groups small.

Audience and Teaching

The protégé effect. Catherine Chase, Doris Chin, Marily Oppezzo, and Daniel Schwartz (2009) gave eighth graders biology software with a twist. Some were told they were teaching a computer character, a teachable agent that reasoned from whatever they taught it. Others used the same software to learn for themselves. The teachers spent more time on the learning activities and learned more, and the benefits were largest for lower-achieving students.

Expecting to teach. People who studied a passage expecting to teach it to another student, though none actually did, recalled more of it, organized it better, and answered more questions about its main points than people expecting a test (John Nestojko and colleagues, 2014). A meta-analysis found that preparing to teach had a small-to-medium benefit, and a medium benefit when actual teaching followed, including on delayed tests (Keiichi Kobayashi, 2019). You may know this from your own life: read an article for a work report and you skim for facts; read it planning to explain it at dinner and you find the story.

The presence of others cuts both ways. Research on social facilitation finds that an audience improves performance on well-learned tasks and impairs it on new and difficult ones (Robert Zajonc, 1965). A driver who has parallel parked for ten years may slide in perfectly while a café full of people watches. A sixteen-year-old in the same spot will clip the curb. An audience helps Practice near mastery, and can hurt learning at its most fragile.

When it points at the object. An audience that needs the work, the student being taught, the council reading the letter, points at the object and raises its worth.[8] An audience that judges the performer points at the performer.

With age. Young children show and tell freely. Sensitivity to being watched by peers is especially strong in adolescence, which makes the kind of audience matter most then.

Design moves. Have learners teach: a younger child, a parent, a teachable agent. Give finished work a real audience. Keep new, fragile learning private until it's ready.

Imitation and Prestige

Prestige bias. Maciej Chudek, Sarah Heller, Susan Birch, and Joseph Henrich (2012) had three- and four-year-olds watch two adults. Bystanders watched one of them attentively and ignored the other. The children were more than twice as likely to learn from the adult the bystanders had watched, and the

effect was specific to the kind of activity that adult had been doing. Even at three, children watch whom the room watches.

Children copy even what doesn't matter. Show a child how to get a toy out of a clear puzzle box, and include a few pointless steps, such as tapping the top with a stick before opening the door. A chimpanzee watching the same demonstration skips the tapping and opens the door. Children keep the needless steps, even when told they can leave them out (Derek Lyons, Andrew Young, and Frank Keil, 2007). This is **overimitation**, and it appears across very different cultures. Children read an adult's deliberate actions as meaningful: if a capable adult does it, there's probably a reason, even an invisible one. It looks inefficient, and it's also how a culture passes on its tools and rituals intact, long before the learner understands why each step matters.

When it points at the object. Children copy more than what adults do. They copy where adults look and what adults seem to care about. A model absorbed in the work passes on attention to the work. A model admired for status passes on desire for status, as the box on mimetic desire describes. A manager who gives speeches about quality while visibly chasing her own promotion will find her team copying the chase.

With age. Faithful copying emerges between eighteen months and three years and grows stronger through childhood. Adults do it too.

Design moves. Let children watch adults do real work with visible care. Explain why each step matters, so children copy purposes as well as motions. Offer models slightly ahead as well as far ahead.

Belonging

Minimal connections move effort. Adults who learned they shared a birthday with a student majoring in math persisted longer on an unsolvable math puzzle (Walton, Geoffrey Cohen, and colleagues, 2012). Four-year-olds told they belonged to a "puzzles group" persisted longer on a hard puzzle than children told they were a "puzzles child" (Allison Master and Walton, 2013). A shared birthday with a stranger, or a single word changing "child" to "group," was enough.

Belonging at scale. In a randomized trial with 92 first-year college students, a brief exercise framing social adversity as common and temporary (everyone feels out of place at first; it passes) raised African American students' grades over three years and halved the racial achievement gap (Walton and Cohen, 2011). A replication with 26,911 students at 22 institutions found that the online version, under thirty minutes long, raised the rate of completing the first year full-time, especially for groups that had historically progressed at lower rates. It worked only where students' groups had real opportunities to belong (Walton and colleagues, 2023). The exercise could clear away the feeling of not belonging. It couldn't create a place to belong where none existed.

When it points at the object. Belonging to a group defined by the work, people who build things, points at the object. Belonging that depends on standing, fitting in by scoring well, points at standing.

With age. Belonging matters at every age and weighs most at transitions: a new school, a new team, the first year away. Identity labels, a close relative, are covered in Chapter 8.[9]

Design moves. Make clear that struggle is normal and shared. Form groups around the work. Give newcomers real roles from the start.

Praise

Praise the work. In a classic experiment, fifth graders who succeeded at a set of problems were praised either for their intelligence ("you must be smart at this") or for their effort ("you must have worked hard"). Those praised for intelligence later avoided challenging problems, enjoyed the task less, and performed worse after a failure (Claudia Mueller and Carol Dweck, 1998). In a study of families at home, parents' praise of their toddlers' effort between 14 and 38 months predicted children's belief at ages seven and eight that ability can grow; praise of the child's traits wasn't linked to the opposite belief (Elizabeth Gunderson and colleagues, 2013).

The evidence is weaker than its fame. A close replication with 624 Chinese children aged nine to thirteen found the praise manipulation had no

effect on any of the eight motivation and attribution measures the original study used, and only a marginal effect on performance after failure (Yue Li and Timothy Bates, 2019). Research on growth-mindset interventions more broadly has found small average effects.

Inflated praise backfires for some. Adults give more inflated praise ("that's incredibly beautiful!") to children with low self-esteem, trying to lift them. In an experiment, inflated praise led those children to avoid challenges, while it had the opposite effect on children with high self-esteem (Eddie Brummelman and colleagues, 2014). Inflated praise sets an insecure child on a high pedestal, and the safest thing to do on a pedestal is stand still. Praise that describes the work builds a staircase instead: each step is solid, and a slip costs one step.

When it points at the object. Praise that describes the work, what the learner did, how they approached it, what changed, is feedback about the object. Praise that points at the person ("you're so smart") is a label, and labels can backfire on hard tasks.[10]

With age. Praise habits start early. What parents praised in toddlerhood predicted children's beliefs about ability years later.

Design moves. Describe what the learner did. Skip superlatives. Praise strategies and choices. Let the work's own result do most of the praising.

The Relationship With the Teacher

Warmth goes with engagement. A meta-analysis of 99 studies from preschool through high school found that warm teacher-student relationships were associated with students' engagement at medium-to-large strength, and with achievement at small-to-medium strength. Negative relationships showed the reverse (Debora Roorda and colleagues, 2011). A follow-up found the link to achievement ran partly through engagement (2017). These studies are correlational; engaged students may also draw warmer relationships.

Wise feedback. David Yeager and his colleagues (2014) gave middle school students critical comments on their essays, with one of two notes attached. The neutral note said the comments were there to give feedback.

The other said: I'm giving you these comments because I have very high expectations, and I know you can reach them. Among African American students, 64 percent revised their essays after the second note, against 27 percent after the neutral one. High standards and belief in the learner, stated together, made criticism usable.

When it points at the object. Wanting to impress someone you care about is among the strongest outside drivers there is. It points at the object when the admired adult cares visibly about the work, and at approval when the adult cares mainly about the child's performance. It's the engine of the tutoring model in Chapter 13.[11]

With age. The associations were stronger in the higher grades, when relationships with teachers usually grow thinner.

Design moves. Hold high standards and say why. Show your own interest in the work. Know each learner well enough to read them.

❧ Story Drivers

Narrative

A textbook can feel like chewing sand. A thriller will keep you up until three in the morning on a work night.

Raymond Mar, Jingyuan Li, Anh Nguyen, and Cindy Ta (2021) pooled more than 75 samples and 33,000 participants and found that narrative texts were more easily understood and better recalled than expository ones, and the result held across studies and conditions. In gamification research, adding **game fiction**, a story the learner plays inside, was among the features that most improved learners' behavior (Michael Sailer and Lisa Homner, 2020).

When it points at the object. A story about the object carries the learner into the material: the dig team that has to measure a femur to learn how big the dinosaur was. A story laid on top of the material carries the learner away from it: the cartoon between drills, where the math is the toll you pay to reach the story. Egan's tools for making material the learner's own are collected in Chapter 8.[12]

With age. Story is the strongest tool of early childhood, and the advantage of narrative holds for adults as well.

Design moves. Build the story from the material's own history: who discovered it, what they were trying to solve, what went wrong. The real history of chemistry, with its eccentrics poisoning themselves and its missed discoveries, beats any invented space alien. Cast the learner in a role inside the problem. Let the story end in the material, rather than in a reward.

Curiosity and Mystery

The information gap. George Loewenstein (1994) proposed that curiosity arises when people notice a gap between what they know and what they want to know: an itch.

Curiosity helps memory. Matthias Gruber, Bernard Gelman, and Charan Ranganath (2014) had people read trivia questions in a brain scanner and rate how curious they were about each answer. People remembered the answers they'd been curious about better, both immediately and a day later, and curiosity recruited the same brain reward circuits that respond to tangible rewards (see also Min Jeong Kang and colleagues, 2009). Closing a gap you care about registers, in the brain, something like being paid. The same study found better memory even for unrelated faces shown during high-curiosity moments, though later studies suggest that spillover is weaker and less consistent.

Curiosity peaks at partial knowledge. In Kang's study, curiosity was highest when people were moderately confident of an answer. Knowing nothing gives the itch nothing to grip. Knowing everything leaves no itch. Knowing a little makes you want the rest.

School quiets questions. Barbara Tizard and Martin Hughes (1984) recorded 30 four-year-old girls at home and at nursery school. At home they asked about 26 questions an hour, often in long chains on a single topic: why, and then why that, and then what about this. At nursery school they asked about 2 an hour, and that kind of sustained exchange almost never happened. The children were the same children. The setting changed what they voiced.

When it points at the object. A gap inside the object leads into the object: why does the maple turn red weeks before the oak? A gap built to be clicked, the cliffhanger that resolves into nothing, is capture wearing curiosity's clothes.

With age. Children don't stop wondering when they start school. The capacity is there at every age.

Design moves. Start with a question the learner half knows. Leave the gap open long enough to be felt. Answer a question with the next question. Make room for chains of questions on one topic.

Play and Make-Believe

The Batman effect. Rachel White and her colleagues (2017) gave four- and six-year-olds a dull, repetitive computer task for ten minutes, with an attractive video game on a tablet beside them that they could switch to whenever they liked. Some children were asked to think about how they were doing in their own voice ("Am I working hard?"). Some were asked to refer to themselves in the third person ("Is Jamie working hard?"). Some were given a costume prop and asked to pretend to be a hard-working character such as Batman ("Is Batman working hard?"). The pretending children spent the most time working, more than the third-person group, and far more than the children who stayed in their own perspective. A child may want the tablet. Batman has a job to do. Play can carry Practice, and it connects to the self-control research in Chapter 8.[13] (If you ever need to get through your taxes, consider a cape.)

Play-based curricula: mixed evidence. Tools of the Mind, a preschool curriculum built on structured make-believe, improved executive function in early trials (Adele Diamond and colleagues, 2007) and in a kindergarten trial with 759 children (Clancy Blair and Cybele Raver, 2014). A larger trial with 877 preschoolers in 60 schools found no benefits and possible harms (Dale Farran and Sandra Wilson, 2014). Positive effects appeared most often in high-poverty settings.

What play hasn't been shown to do. A major review found the evidence insufficient to show that pretend play is crucial to development; it may be one route among many (Angeline Lillard and colleagues, 2013). Play works as a driver. Claims that it's necessary for development remain unproven.

When it points at the object. Play inside the object carries the object: a pretend shop where making change takes real arithmetic. Play beside it, a game of tag as a reward for finishing the worksheet, leaves the object behind.

With age. Make-believe is strongest in early childhood. Later, games with rules and role-play take its place.

Design moves. Give children roles that require the skill. Use pretend to carry dull reps. Keep real materials inside the play.

Entertainment

The seductive details effect. Afraid of boring an audience, most of us decorate: a joke about the cat, a funny image, a story from the weekend. Narmada Sundararajan and Olusola Adesope (2020) pooled the research on interesting but irrelevant material and found a small-to-moderate negative effect on recall and transfer (see also Günter Daniel Rey, 2012). Working memory is small. Spend some of it on the cat, and on working out what the cat has to do with chemical bonds, and there's less left for the bonds. You've probably sat through a charming, funny presentation and walked out unable to say what the new policy was.

Entertainment built around the material works. Children who had access to *Sesame Street* when it first aired in 1969, because of where they happened to live, were more likely to be at the right grade level for their age later on, especially boys (Melissa Kearney and Phillip Levine, 2019).

Humor. Reviews of classroom humor find its effects depend on the kind. Humor about the material tends to help; humor unrelated to it, or at students' expense, doesn't (John Banas and colleagues, 2011).

When it points at the object. The child from Chapter 4, shouting with glee as her rocket hits a record height, is entertained by the object itself.

Entertainment about the object is engagement. Entertainment about something else is capture.

Design moves. Make the material itself the source of delight. Cut decorative jokes and images from hard lessons. Save pure entertainment for rest.

❧ Structural Drivers

Structural drivers are the machinery of school and work: rewards, grades, money, progress bars, streaks, rankings, deadlines. They're the easiest drivers to install and the easiest to count, which is why they're everywhere.

Rewards

Ask a friend to help you move a couch on a Saturday and they'll grumble and come. Buy them a pizza afterward and the favor is sealed. Offer them $4.50 for the afternoon and they'll be insulted. The money turns a favor into a job, and a badly paid one.

What the research shows. A meta-analysis of 128 experiments found that expected, tangible rewards reduced free-choice intrinsic motivation, while informational positive feedback increased it. Unexpected rewards, given after the fact, didn't reduce motivation (Edward Deci, Richard Koestner, and Richard Ryan, 1999). The pizza after the couch is safe. The size of the effect remains debated. The preschoolers who drew less after being promised a certificate for drawing are in Chapter 8.[14]

What rewards buy. Incentives predict the quantity of performance better than its quality (Cerasoli and colleagues, 2014), and acting for rewards isn't associated with persistence (Howard and colleagues, 2021). Rewards buy more of the work. They rarely buy better work, or work that lasts after the payments stop.

When it points at the object. Feedback that tells a learner how they're doing points at the object. A prize for doing it points at the prize.

With age. In the same meta-analysis, the undermining effect of tangible rewards was larger for children than for college students.

Design moves. Save rewards for tasks with no interest to lose, and keep them away from work the learner already enjoys. Make rewards unexpected and informational. Phase them out as the work takes over.

Grades

Comments beat grades. Ruth Butler (1988) randomly assigned twelve fifth- and sixth-grade classes to receive individual comments, numerical grades, or both after interesting tasks. Interest and performance were highest after comments alone, for high and low achievers alike, and a grade given alongside a comment wiped out the comment's benefit. Later studies haven't always found grades this damaging, and the effects vary widely.

Hand a student a paper with three paragraphs of careful comments and a B+ at the top, and watch where the eyes go. (To the B+, and then to the neighbor's paper.)

Removing the rank. When Mayo Medical School moved its first-year courses to pass/fail, you might have expected the future doctors to coast. Students reported less stress and greater group cohesion, and their scores on the national board exam didn't change (Rohe and colleagues, 2006). A review of such changes found improvements in well-being across studies (Spring and colleagues, 2011).

Why. A comment points at the work: what to change next. A grade points at a rank. Given both, learners look at the grade.

The classroom's default driver. Grades are the outside driver the classroom form runs on, which is part of why that form struggles to produce ownership.[15]

With age. Grades arrive with schooling and gain weight every year, peaking in the competition for college.

Design moves. Comments first. Grades rarely, and never alongside comments. Pass/fail or mastery standards wherever ranking serves no purpose.

Money: Inputs and Outputs

Paying for results did little. Roland Fryer (2011) ran randomized trials in more than 200 urban schools. Students in Dallas were paid to read books ($2 a book, confirmed by a short quiz), in New York for test performance, and in Chicago for grades. Effects on state test scores were statistically zero in each city. The only significant effect was for English-speaking students in Dallas, the group paid to read. Fryer's interviews suggested why: students were excited by the money and didn't know how to turn effort into higher scores.

Offer me a million dollars to play a Mozart concerto tomorrow and you'll get my full attention and no Mozart. I can't play the piano. Motivation without capacity produces a lot of staring at the keys.

Immediacy is everything. In field experiments with thousands of students around Chicago, rewards announced just before a low-stakes test raised scores, and the effect vanished entirely when the reward was delayed. Rewards framed as losses (here's the money; you keep it if you improve) did somewhat better (Steven Levitt, John List, Susanne Neckermann, and Sally Sadoff, 2016).

When it points at the object. Paying for an act that is itself contact with the object, reading a book, can help. Paying for an outcome points at the payout. The test-day rewards changed effort on the test and left learning where it was. They raised the measure, which is what capture does.

With age. Trophies worked about as well as cash for younger students and less well for older ones (Levitt and colleagues, 2016).

Design moves. If you pay at all, pay for contact with the object, and pay at once. Never pay for outcomes learners don't know how to produce.

Visible Progress and Feedback

Progress in meaningful work. Teresa Amabile and Steven Kramer (2011) collected nearly 12,000 daily diary entries from 238 professionals on 26 project teams in seven companies. The single strongest booster of motivation and positive emotion at work was making progress in meaningful work, even small progress.

Feedback helps, except when it turns to the self. Avraham Kluger and Angelo DeNisi (1996) pooled 607 effects and found that feedback improved performance on average (d = 0.41). More than a third of feedback interventions made performance worse. Feedback helped less the more it drew attention toward the self and away from the task.

Picture a tennis coach. "Tilt your wrist down a little as you swing" gives the player something to do on the next ball. "You're just not very coordinated" gives her something to think about instead, and the thing is herself. So does "you're in the bottom fifth of the group," however gently it's said. Kluger and DeNisi's finding is this chapter's test in its most general form.

When it points at the object. Progress that reports movement on the object points at the object: a first loaf that rises, a passage that finally holds at tempo. Points that climb whatever the learning does point at the points. A head start that already feels like the learner's own is covered in Chapter 8.[16]

With age. Young children need progress they can see and touch. Older learners can track progress in the abstract.

Design moves. Show progress on the object itself. Describe the next step, and leave the person out of it. Keep progress private where comparison would hurt.

Streaks

The little flame with a 45 beside it does get you to open the language app. Then you travel, miss a day, lose the 45, and don't open the app again for six months.

The streak becomes the goal. Across seven studies, highlighting an intact streak in a behavior log increased people's next engagement in the behavior, compared with highlighting a broken one. The effect depended only on how the log represented past behavior, and the behavior itself didn't matter, because people come to treat keeping the streak as a goal in its own right. The damage from a broken streak was larger when people blamed themselves for the break, and smaller when they could repair it (Joseph Silverman and

Alixandra Barasch, 2023). Tap through one lesson at 11:55 p.m. to keep the flame alive, and Spanish has become incidental to the chain.

Habits survive a miss. Phillippa Lally and her colleagues (2010) followed 96 people forming a new daily habit. The behavior took a median of 66 days to become as automatic as it would get (well past the 21 days of self-help lore), with a range of 18 to 254 days. Missing a single day didn't materially affect the process. The habit is more forgiving than the streak. Forty-four days out of forty-five is a fine record.

When it points at the object. A streak that counts the work, pages read or pieces played, stays near the object. A streak that counts check-ins can replace the object as the thing being pursued, and a single break can end both.

Design moves. Count the work, and leave logins out of it. Allow repair. Emphasize the long pattern over the unbroken chain.

Leaderboards

Leaderboards work like goals. In an experiment, adding a leaderboard to a task raised performance about as much as assigning a difficult goal. People implicitly aimed near the top of the board, and the effect depended on how committed they were to that goal (Richard Landers, Kristina Bauer, and Rachel Callan, 2017).

Comparison has costs. In a virtual classroom experiment with 80 women taking a math test, leaderboards shifted academic identification and, in some conditions, lowered math performance. The harm ran through social comparison more than through stereotype threat (Katheryn Christy and Jesse Fox, 2014).

When it points at the object. A leaderboard is other-referenced competition made permanent and public. It works best for the learners near the top, and it points everyone else at the rank. A woman who jogs twice a week gains nothing from learning she's 10,412th in the app, somewhere behind an Olympian. Show her she's neck and neck with the neighbor who also jogs twice a week, and she laces up.

Design moves. Replace public rankings with personal goals. If a board is used, show learners their near neighbors instead of the whole field, and change what it measures often.

Deadlines

You may be thinking that deadlines are the only reason anything gets done. Without one, the report waits and the evening goes to television.

Pressure and interest. In a classic experiment, people working under a deadline later showed less interest in the task than people working without one (Teresa Amabile, William DeJong, and Mark Lepper, 1976).

A caution about the evidence. The best-known study showing that deadlines improve performance (Dan Ariely and Klaus Wertenbroch, 2002) has been retracted, and a 2026 replication didn't reproduce its results.

Plans help more than pressure. A meta-analysis found that forming implementation intentions, specific if-then plans for when and where to act, had a medium-to-large effect on reaching goals (Peter Gollwitzer and Paschal Sheeran, 2006). "Finish the report by Friday" is a deadline. "Tuesday at eight, coffee poured, I sit down and write the first five hundred words" is a plan, and it tells you what to do when Tuesday at eight arrives.

When it points at the object. A deadline tied to a real event points at the object: opening night, the science fair, the morning the family leaves on the trip. An arbitrary Friday at five points at the calendar.

Design moves. Let learners set dates, with an adult checking in. Tie deadlines to real events. Pair them with specific plans.

Novelty and the Gamification Bundle

Gamification helps on average. A meta-analysis of 40 experiments found that gamification improved cognitive ($g = 0.49$), motivational ($g = 0.36$), and behavioral ($g = 0.25$) learning outcomes. Game fiction, and competition combined with collaboration, were among the most effective elements (Sailer and Homner, 2020). Well-designed digital games beat non-game instruction

(g = 0.33), and enhanced game designs added further gains (Douglas Clark, Emily Tanner-Smith, and Stephen Killingsworth, 2016).

Novelty fades. In one of the few semester-long studies, one course used leaderboards and badges and a matched course didn't. Over sixteen weeks, students in the gamified course grew less motivated, less satisfied, and less empowered, and their lower motivation carried through to lower final exam scores (Michael Hanus and Jesse Fox, 2015). The badges that sparkled in September were a chore by December.

Games and gamification are different things. In a good learning game, the mechanics are the object: the player has to understand proportion to win. In most gamification, points and badges are laid on top of an unchanged task. The first points at the object. The second points at itself.

Design moves. Prefer game fiction and team competition to points and badges. Build the mechanics from the material. Plan for the novelty to fade.

Outside Drivers in Inquiry and Practice

The three families fit the two systems differently. What follows is the book's hypothesis. The evidence is consistent with it, and none of it was designed to test it.

Structure suits Practice. Practice is repetition toward a standard, and repetition is where attention flags: the fortieth scale, the hundredth free throw, the vocabulary list on the fridge. Streaks that count the work, visible progress, and competition against your own record help carry the reps. The Batman effect was measured on exactly this kind of task: dull, repetitive, and worth finishing.

Story suits Inquiry. Inquiry runs on a question the learner owns, and curiosity is already its own reward. Story and mystery can open a question: the femur on the dig, the maple that turns before the oak. Rewards and rankings sit poorly with it, because the undermining research finds the strongest damage where the activity was interesting to begin with. Pay a child

to solve a mystery she's already desperate to solve, and the next mystery may need paying for too.

Social drivers serve both. Cooperation, an audience, and admired models can carry a question or a rep. The four-year-old at the ant trail follows the ants on her own, and she follows them longer with a parent crouched beside her asking where they're going. A choir carries a singer through the twentieth run of a hard bar.

	Fits well	Use with care
Practice	Streaks that count the work, visible progress, competing with your own record, play and pretend, teams	Arbitrary deadlines, public rankings
Inquiry	Story, mystery, partial knowledge, an audience that needs the answer, admired models	Expected rewards, grades, leaderboards
Both	Cooperation, teaching others, belonging to a group defined by the work, a teacher who cares about the work	Praise of the person

Most real learning runs both systems, often in the same hour, so most real settings mix the drivers. A science fair project is Inquiry while the question is open and Practice when the poster has to be lettered and the talk rehearsed. The driver can change when the system does.

The design consequence. Reach for structure to carry Practice, for story and mystery to open Inquiry, and for other people to carry both. Be most careful with rewards around Inquiry.

❧ Working Tests

Five questions tell capture from engagement in the room.

1. **The removal test.** Remove the driver. Does the learner keep going? The rewards research measures exactly this: what learners choose to do once the reward stops. The reading chart came down on the first of September, and the test was over by October.
2. **The attention test.** Where does the learner look: at the object, or at the count, the rank, the prize? Watch the eyes on the paper with the B+.

3. **The ownership test.** Is the driver moving the learner toward owning the work, or standing in for ownership?
4. **The quality test.** Is the work getting better, or only more plentiful? Forty thin books is a lot of books.
5. **The loss test.** What happens after a loss: a broken streak, a lost race, a missed badge? A chess player who loses a game she cares about sets the pieces back up to find the move where it went wrong. A learner held by the streak alone closes the app. Engagement survives a loss and often sharpens. Capture collapses.

No single test settles a case. Two or three pointing the same way usually do.

❧ Scaffolding: Using Outside Drivers Well

Scaffolding on a building goes up first and comes down when the walls can stand. Outside drivers work the same way.

Start strong, then step back. Outside drivers are most useful at the start, before the core drivers of ownership, fit, fidelity, and worth have taken hold.[17] They should fade as those drivers do the work. A streak may get you out the door for your first month of running. By the third month, the run should be doing the pulling.

Point every driver at the object. Competition against your own record or another team. Stories set inside the material. Progress that reports real movement. Streaks that count the work and can be repaired. Comments in place of grades, and feedback in place of prizes.

Watch who is harmed. The same competition or leaderboard that lifts one learner can sink another. The speller at the top of the bracket and the speller knocked out in the first round sat through the same bee. Reading engagement shows which is happening.[18]

Mind the direction. The same tools power manipulative design. Streaks, progress bars, and near-neighbor boards were refined in apps built to hold attention for its own sake. A driver that leads toward the learner's own

ownership serves the learner. One that keeps them hooked serves someone else.[19]

The Long Run

The evidence is mostly short. Most studies of outside drivers measure days or weeks. The few longer ones, like Hanus and Fox's semester, find effects fading or reversing as the novelty wears off.

The book's prediction. Drivers that point away from the object should raise short-term measures (effort, time on task, scores) while weakening retention and attachment later. Drivers that point at the object should compound, because they hand the learner over to the object's own pull.[20] The prediction is testable: follow learners for a year or more, and measure what they do when the drivers stop.

Much of what you use every day runs on short-term structural drivers: the badges in the app, the stamps on the coffee card, the quarterly ranking at work, the due date on Friday. If the prediction holds, those systems are trading later curiosity for compliance now, and nobody is keeping the books on the trade.

The classroom's arrangement has also left the school building. A feed puts you among peers wherever you are, ranked by likes, views, and followers, which are grades and badges by other names. In that room the photograph and the essay can vanish into the rivalry the way the subject does in a classroom, and the work becomes a way to outrank the next account. Girard would recognize it as internal mediation with no closing bell.

Next June the library will post its chart again. It could count pages instead of books, pin the children's own recommendations on the wall for other children to find, and skip the pizza. The girl might read fewer books in July. She might also still be reading in October, like her brother, who is on the third volume now.

Open Questions

- Which outside drivers most often turn capture into engagement, and how long does the handover take?
- Can a driver be withdrawn gradually without triggering the collapse a broken streak causes?
- Why does competition affect girls and boys differently at such young ages, and what forms of competition avoid it?
- Do story drivers help Inquiry at every age, or mainly while story is the learner's strongest tool?
- Does the long-run prediction hold: short-term gains from drivers that point away from the object, and weaker retention later?
- Does a school day spent among same-age rivals help explain bullying, as Girard's account of scapegoating would predict?

If you've watched an outside driver hand a learner over to the work, or watched one fail to, I'd like to hear about it.

Notes

1. See Chapter 4, “What Drives Engagement.”
2. See Chapter 4, “What Engagement Is.”
3. See Chapter 8.
4. See Chapter 8.
5. See Chapter 2, “The Form Is the Cause.”
6. See Chapter 7, “The Early Drive to Matter”; Chapter 8, “Handing Over Ownership”; and Chapter 13.
7. See Chapter 8, “Scales of Ownership and Modes of Agency.”
8. See Chapter 7.
9. See Chapter 8, “Seeding Ownership.”
10. See Chapter 8, “Seeding Ownership.”
11. See Chapter 13.

12. See Chapter 8, "Seeding Ownership."
13. See Chapter 8, "Passivity Is the Default."
14. See Chapter 8, "Driver or Passenger."
15. See Chapter 2, "The Form Is the Cause."
16. See Chapter 8, "Seeding Ownership."
17. See Chapter 4, "What Drives Engagement."
18. See Chapter 4, "Reading Engagement: Zoom and Aperture" and "Proxies for Engagement."
19. See Chapter 8, "Seeding Ownership."
20. See Chapter 5, "Desirable Difficulties."

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